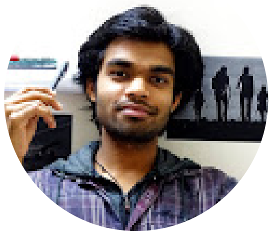

# Akshay Gaikwad

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**Research Profiles:** Google Scholar, ResearchGate
**Homepage:** https://agtomo.github.io

## Education

**Staff Scientist** — Aug. 2026 – Present
Applied Quantum Physics Laboratory division — **Chalmers University of Technology, Sweden**

**Postdoc** | **Characterization of quantum processors** — Jan. 2024 – May. 2026
Supervisors: Prof. Anton Frisk Kockum — **Chalmers University of Technology, Sweden**

**Research Assistant** — May. 2023 – Dec. 2023
Supervisors: Prof. Kavita Dorai and Prof. Arvind, Dept.of Physical Sciences — **IISER Mohali, India**

**PhD, Physics** | **Quantum simulation, tomography and NMRQC** — Aug. 2016 – May. 2023
Supervisors: Prof. Kavita Dorai and Prof. Arvind, Dept.of Physical Sciences — **IISER Mohali, India**

**BS-MS Integrated Dual Degree** | ***CPI: 7.8/10, Major: Physics*** — Aug. 2011 – Aug. 2016
MS Thesis: Characterizing quantum processes on NMR — **IISER Mohali, India**

**Senior Secondary Education** | ***Percentage: 83 (88 in Basic Science)*** — Aug. 2009 – Aug. 2011
Subjects: Physics, Chemistry, Biology, Mathematics, English, Hindi — S.N.B.P. Jr. College, Pune

**Secondary Education** | ***Percentage: 90.92*** — Aug. 2009
Subjects: Science, Mathematics, Social Science, Hindi, English, Marathi — BPHS, Satara

## Research Interests

- Quantum characterization, verification and validation (QCVV) protocols
- Quantum tomography and certification
- Classical data processing and optimization methods
- Machine learning methods for quantum technology
- Linear algebra for quantum computing
- Quantum simulation and algorithms
- Superconducting qubits and nuclear spin qubits

## Preprints

2025 **Akshay Gaikwad**, Manuel Sebastian Torres and Anton Frisk Kockum.
*Quantum measurement tomography with mini-batch stochastic gradient descent*
arXiv:2511.15682 (2025)

2024 Gayatri Singh, **Akshay Gaikwad**, Arvind and Kavita Dorai.
*Experimental decoherence mitigation using a weak measurement-based scheme and the duality quantum algorithm*
arXiv:2409.12752 (2024)

2024 **Akshay Gaikwad**.
*Novel techniques for efficient quantum state tomography and quantum process tomography and their experimental implementation*
arXiv:2401.09941 (2024)

## Published articles

2026 Aniket Patel, **Akshay Gaikwad**, Tangyou Huang, Anton Frisk Kockum and Tahereh Abad.
*Selective and efficient quantum state tomography for multi-qubit systems*
Physical Review Research 8, 013339 (2026)

2025 Tangyou Huang, **Akshay Gaikwad**, Ilya Moskalenko, Anuj Aggarwal, Tahereh Abad, Marko Kuzmanovic, Yu-Han Chang, Ognjen Stanisavljevic, Emil Hogedal, Christopher Warren, Irshad Ahmad, Janka Biznárová, Amr Osman, Mamta Dahiya, Marcus Rommel, Anita Fadavi Rousari, Andreas Nylander, Liangyu Chen, Jonas Bylander, Gheorghe Sorin Paraoanu, Anton Frisk Kockum and Giovanna Tancredi.
*Quantum Process Tomography with Digital Twins of Error Matrices*
Physical Review Letters 135, 230601 (2025)

2025 **Akshay Gaikwad**, Manuel Sebastian Torres, Shahnawaz Ahmed and Anton Frisk Kockum.
*Gradient-descent methods for fast quantum state tomography*
Quantum Science and Technology 10 045055 (2025)

2025 Christian Križan, Janka Biznárová, Liangyu Chen, Emil Hogedal, Amr Osman, Christopher W. Warren, Sandoko Kosen, Hang-Xi Li, Tahereh Abad, Anuj Aggarwal, Marco Caputo, Jorge Fernández-Pendás, **Akshay Gaikwad**, Leif Grönberg, Andreas Nylander, Robert Rehammar, Marcus Rommel, Olga I. Yuzephovich, Anton Frisk Kockum, Joonas Govenius, Giovanna Tancredi and Jonas Bylander.
*Quantum SWAP gate realized with CZ and iSWAP gates in a superconducting architecture*
New Journal of Physics 27 074507 (2025)

2025 Jiaying Yang, Maryam Khanahmadi, Ingrid Strandberg, **Akshay Gaikwad**, Claudia Castillo-Moreno, Anton Frisk Kockum, Muhammad Asad Ullah, Gäran Johansson, Axel Martin Eriksson and Simone Gasparinetti.
*Deterministic generation of frequency-bin-encoded microwave photons*
Physical Review Letters 134, 240803 (2025)

2025 Jiaying Yang, Ingrid Strandberg, Alejandro Vivas-Viana, **Akshay Gaikwad**, Claudia Castillo-Moreno, Anton Frisk Kockum, Muhammad Asad Ullah, Carlos Sanchez Munoz, Axel Martin Eriksson and Simone Gasparinetti.
*Entanglement of photonic modes from a continuously driven two-level system*
Npj Quantum Information 11, 69 (2025)

2024 **Akshay Gaikwad**, Omkar Bihani, Arvind and Kavita Dorai.
*Neural network assisted quantum state and process tomography using limited data sets*
Physical Review A 109, 012402 (2024)

2023 **Akshay Gaikwad**, Gayatri Singh, Kavita Dorai and Arvind.
*Direct tomography of quantum states and processes via weak measurements of Pauli spin operators*
The European Physical Journal D 77: 209 (2023)

2022 **Akshay Gaikwad**, Krishna Shende, Arvind and Kavita Dorai.
*Implementing efficient selective quantum process tomography of superconducting quantum gates on IBM quantum experience*
Scientific reports 12 (1), 1-11 (2022)

2022 **Akshay Gaikwad**, Arvind and Kavita Dorai.
*Experimental simulation of open quantum dynamics using Sz-Nagy's dilation algorithm using NMR*
Physical Review A 106, 022424 (2022)

2022 **Akshay Gaikwad**, Arvind and Kavita Dorai.
*Efficient characterization of quantum processes from reduced data set via compressed sensing using NMR*
Quantum Information Processing 21, (12) (2022)

2021 **Akshay Gaikwad**, Arvind and Kavita Dorai.
*True experimental reconstruction of quantum states and processes via convex optimization using NMR*
Quantum Information Processing 20, 19 (2021)

2020 **Akshay Gaikwad**, Krishna shende and Kavita Dorai.
*Experimental demonstration of optimized quantum process tomography on the IBM quantum experience*
International Journal of Quantum Information 19 (07), 2040004 (2020)

2018 **Akshay Gaikwad**, Diksha Rehal, Amandeep Singh, Arvind and Kavita Dorai.
*Experimental demonstration of selective quantum process tomography on NMR*
Physical Review A 97, 022311 (2018)

## Peer Reviews

Quantum Information Processing (5 reviews)
Results in Physics (3 reviews)
Physical Review Letters (2 reviews)
New Journal of Physics (2 reviews)
Physical Review Applied (1 review)
Physical Review E (1 review)
Scientific Reports (2 review)

## Conference Participation

2026 Delivered a **talk** at APS Global Physics Summit held at Denver, CO, USA from March 16–20, 2025

2025 Presented a **poster** at Assessing Performance of Quantum Computers (APQC) workshop held at Estes Park, CO, USA from September 21-25, 2025

2025 Delivered a **talk** at APS Global Physics Summit held at Anaheim, CA, USA from March 16–21, 2025

2025 **Attended** Nordic Workshop on Continuous Variable Quantum Technology, held at Chalmers University of Technology, Gothenburg, Sweden on February 24, 2025

2024 Presented a **poster** at Assessing Performance of Quantum Computers (APQC) workshop held at Estes Park, CO, USA from October 7-10, 2024

2024 **Attended** CodeRefinery workshop at Chalmers University of Technology, Gothenburg, Sweden from August 27-29, 2024

2024 **Attended** Nordic Workshop on Continuous Variable Quantum Technology, held at KTH Royal Institute of Technology, Stockholm, Sweden from May 27-28, 2024

2023 **Attended** Young Quantum-2023 (YouQu-2023) conference, held at Harish-Chandra Research Institute (HRI), Allahabad, India from February 15-18, 2023

2023 **Attended** International Conference on Quantum Computing and Communications (QCC2023), held at Baba Farid College Bathinda, Punjab, India from February 9-11, 2023

2020 **Attended** Young Quantum - 2020 (YouQu-2020) online conference, held at Harish-Chandra Research Institute (HRI), Allahabad, India from October 12-15, 2020

2020 **Attended** International conference on *Quantum Foundations, Technologies and Applications* (QFTA), held at IISER Mohali, India from December 4-9, 2020

2019 Delivered a **talk** at International Conference on *Quantum Foundations, Technologies and Applications* (QFTA), held at IISER Mohali, India from October 18-21, 2019

2018 Presented a **poster** at International Conference on *Quantum Frontiers & Fundamentals* (QFF), held at Raman Research Institute (RRI) Bengaluru, India from April 30 - May 4, 2018

2018 Presented a **poster** at 24th conference of the *Nuclear Magnetic Resonance Society* (NMRS) of India, held at IISER Mohali, India from February 16-19, 2018

2018 **Attended** *Asia Pacific Conference and Workshop on Quantum Information Science* (APCWQIS), held at IISER Kolkata, India from December 19-23, 2018

2017 Presented a **poster** at 7th *Asia Pacific NMR symposium* (APNMR) and 23rd annual meeting of the *Nuclear Magnetic Resonance Society* (NMRS) of India, held at Indian Institute of Science (IISc) Bangalore, India from February 16-19, 2017

2017 **Attended** *International Conference on Quantum Foundations* (ICQF), held at National Institute of Technology (NIT) Patna, India from December 4-9, 2017

2016 **Attended** *2nd IMSc School on Quantum Information*, held at The Institute of Mathematical Science (IMSc) Chennai, India from December 5-17, 2016

2016 Presented a **poster** at *International Conference on Quantum Foundations* (ICQF), held at National Institute of Technology (NIT) Patna, India from October 17-21, 2016

2016 Presented a **poster** at 22th conference of the *Nuclear Magnetic Resonance Society* (NMRS) of India, held at Indian Institute of Technology (IIT) Kharagpur, India from February 18-21, 2016

## Academic Achievements

2022 Recipient of **senior research fellowship (SRF)** under 'Quantum Information Science and Technology **(QuST)**' research programme initiated by Department of Science and Technology (DST), Government of India.

2018 Recipient of **senior research fellowship (SRF)** at IISER Mohali from 2018-2021.

2016 Qualified **CSIR-UGC National Eligibility Test (NET)** for Lectureship/Assistant Professor in Indian universities and colleges conducted by Council of Scientific and Industrial Research (CSIR), Govt. of India.

2016 Qualified **Graduate Aptitude Test in Engineering (GATE)** organized by Indian Institute of Science (IISc) Bangalore on behalf of the Department of Higher Education, Ministry of Education (MoE), Govt. of India.

2011 Recipient of Innovation in Science Pursuit for Inspired Research **(INSPIRE) Scholarship** valued at Rs.80,000/- per year for five years for higher education from Department of Science and Technology (DST), Govt. Of India.

2011 Qualified **All India Engineering Entrance Examination (AIEEE)** conducted by Central Board of Secondary Education, Delhi (Govt. of India).

2011 Qualified **MHT-CET examination** conducted by Directorates of Medical and Technical Education, Mumbai (Govt. of Maharashtra), India.

2011 Qualified **MANET competitive exam** and got selected in **Maharashtra Academy of Naval Education and Training (MANET)**, Pune for B.Sc. Nautical Science, India.

2011 Stand **among top 1%** students of the board in senior secondary.

2010 Placed **among top 10%** in National Standard Examination in physics, conducted by Indian Association of Physics Teachers.

2008 Received certificate with **first class rank with distinction** in Hindi Creative Writing Exam conducted by Maharashtra State Hindi Teacher Association, Pune, India.

2007 Worked at **Muktangan Exploratory Science Centre**, Pune as part of Vijayadevi Shirke Science Camp (VSSC) and was Representative of my school at (VSSC) conducted by Vijayadevi Shirke Educational Trust, Pune, India.

## Supervision

2026 **Bachelor thesis supervisor**
Esse Brodén (Chalmers University of Technology)
Wiggo Nimvall (Chalmers University of Technology)
Julia Mårtensson (Chalmers University of Technology)
Elsa Selmarker (Chalmers University of Technology)
Fabian Boberg (Chalmers University of Technology)
Alexander Schmeling (Chalmers University of Technology)
**Thesis title**: *Diagnosing errors on a quantum computer*

2025 **Master thesis supervisor**
Sumukh Suresh Moudghalya (Chalmers University of Technology)
**Thesis title:** *Optimizing and scheduling quantum state tomography experiments with graph theory*
Currently doing PhD at University of Rostock in Germany under Prof. Stefan Scheel

2022 **Mentored summer intern**
Omkar Bihani (IISER Mohali, India)
**Project title:** *Neural network assisted quantum tomography*

2020 **Mentored junior PhDs**
Gayatri Singh, Vaishali Gulati and Krishna Shende (IISER Mohali, India)
**Project title:** *General NMR-QC and tomography*

## Other experiences and duties during PhD

- **Tutorial assistant:** TA for undergraduate students for several physics courses and experimental labs, conducting exams, taking viva, checking exam papers and lab files
- **Technical experience:** Hands-on NMR spectroscopy, NMR pulse programming, disassemble and reassemble NMR, liquid nitrogen filling, and flushing in NMR, managing lab website and social media pages
- **Administrative work:** Managing paperwork, bills, and the link between administration and suppliers for the smooth process of fund release and service to the lab.

## Computer and coding Skills

- **Operating systems and softwares:** Linux (Ubuntu, Joli Cloud, Fedora), Microsoft Windows, Fortran, HTML, CSS, ExpEyes, TopSpin, Microsoft Office Suite, LaTex, Matlab, Mathematica, python programming

- **Relevant coding packages:** QuTip, Qiskit, Keras, Pandas, scikit-learn, GRAPE, CVX, QETLAB, YALMIP, UniversalQCompiler, quantikz

## References (for how I performed as a researcher)

- **Anton Frisk Kockum** (Postdoc supervisor)
  Associate Professor, Applied Quantum Physics Laboratory division
  Department of Microtechnology and Nanoscience
  Chalmers University of Technology 41296 Gothenburg, Sweden
  *Email:* anton.frisk.kockum@chalmers.se

- **Kavita Dorai** (PhD supervisor)
  Professor, Department of Physical Sciences
  Indian Institute of Science Education & Research (IISER) Mohali
  Sector 81 SAS Nagar, Manauli PO 140306 Punjab India.
  *Email:* kavita@iisermohali.ac.in

- **Arvind** (PhD supervisor)
  Professor, Department of Physical Sciences
  Indian Institute of Science Education & Research (IISER) Mohali
  Sector 81 SAS Nagar, Manauli PO 140306 Punjab India.
  *Email:* arvind@iisermohali.ac.in

- **Sandeep Kumar Goyal** (Doctoral committee member)
  Assistant Professor, Department of Physical Sciences
  Indian Institute of Science Education & Research (IISER) Mohali
  Sector 81 SAS Nagar, Manauli PO 140306 Punjab India.
  *Email:* skgoyal@iisermohali.ac.in

## References (for how I performed as a supervisor/mentor)

- **Master student**
  Sumukh Suresh Moudghalya
  Email: sumukhm@student.chalmers.se
  Currently, PhD at University of Rostock in Germany under Prof. Stefan Scheel

- **Bachelor students**
  Esse Brodén (esse@student.chalmers.se)
  Wiggo Nimvall (nimvall@student.chalmers.se)
  Julia Mårtensson (juliamar@student.chalmers.se)
  Elsa Selmarker (elsase@chalmers.se)
  Fabian Boberg (bobergf@student.chalmers.se)
  Alexander Schmeling (aleschm@student.chalmers.se)